SUMMER PROJECT WORK

Jayesh Janardan Jadhav
IIT Bombay

Guide: Dr.Subodha Sahoo

HP&SRPD, BARC, Mumbai

1. Goal/Objectives of Internship

- Understanding the properties of materials under extreme conditions of pressure and temperature.
- To have exposure to Raman spectroscopy and DAC-based techniques.
- The material studied during this project is Na2Cu2TeO6.

1.1 Introduction:

As we delve deeper into the cores of planets and stars, we recognize that pressure and temperature escalate. This understanding highlights the pivotal role of high-pressure physics in both geophysics and astrophysics.Remarkably, high-pressure forms of common substances like carbon, boron nitride, and silica (SiO2) emerge as superhard materials synthesizable under extreme conditions.Additionally, zeolites display an intriguing property of expansion under high pressure, positioning them as promising candidates for applications in trapping pollutants and managing nuclear waste.Our primary focus is on exploring phase transitions and associated properties such as superconductivity, which are fundamental to advancing our understanding of materials under extreme conditions. In this study we have picked a Cu based Tellurium oxide system to investigate its high pressure behavior.

2. Experiment and type of equipment used

In this project I have used mainly high pressure Raman spectroscopy and visited to INDUS-II synchrotron facility to have an exposure of high pressure XRD studies. I am explaining below the Raman spectroscopy measurements under pressure in detail.

2.1 Raman spectroscopy

Raman spectroscopy is a powerful technique used to study the vibrational modes of materials, providing insights into their molecular structure, bonding properties. Its core principle relies on the inelastic scattering of light, which distinguishes it from Rayleigh scattering.

Theory and Principle

Raman spectroscopy is based on the interaction of electromagnetic radiation (typically a laser) with the vibrational states of a material.A material exhibits different kinds of vibrational motions depending on its bonding properties and the crystal structure. For a crystalline material, the vibrational motion of all atoms in the unit cell is described in terms of normal modes, which correspond to phonons.When light interacts with a vibrational motion, there is a change in the charge distribution of the atoms in the unit cell, resulting in an interaction with the electromagnetic radiation. This interaction manifests in the form of Raman scattered light.The vibrational modes corresponding to Raman active modes can be observed in both Stokes and anti-Stokes Raman scattering. Stokes scattering involves the photon losing energy to the molecule, while anti-Stokes scattering involves the photon gaining energy from the molecule.

Types of equipment used:Laser, Edge filter, Spectrometer, Diamond anvil cell, gasket, hydrostatic media, Pressure marker.

Laser:

In this project work, we are using a laser with a 532 nm wavelength. As the energy of an EM wave increases, sensitivity increases, but there is a high chance of fluorescence. Low energy comes with low sensitivity but a lower probability of fluorescence.

Spectrometer, filter

In this experiment, we are using a grating-basedspectrometer for light collection and detection. One of the important equipment in the experimental setup is the edge filter. Raman scattered light is weaker by four orders of magnitude compared to the Rayleigh scattered light.So it is hard to detect with so much background noise, hence we use edge filters,which filter out all the light below 532 nm wavelength.

Diamond Anvil Cell (DAC)

The Diamond Anvil Cell (DAC) is a high-pressure device that compresses a sample by placing it in a metallic gasket between a pair of diamond anvils. Due to the smaller area of the anvil tip compared to its overall size, a relatively modest force, such as ~1000 kg, can generate megabar pressures. This mechanism allows for the application of extremely high pressures.

Diamond istransparent across infrared to hard x-rays (5eV to 10 keV), makes them ideal for both x-ray diffraction and spectroscopic studies under pressure. Diamonds must be mounted on backing plates, typically made of hardened steel or tungsten carbide. The top surface of the backing plate is polished smooth, where the diamond is placed, as any non-uniformity can cause the glue between the diamond and the backing plate to flow.

Before applying forces, the diamonds in a DAC must be aligned parallel to each other. One of the backing plates holding a diamond is a "rocker," allowing for tilting via screws, while the other anvil is mounted in a "cylinder" backing plate for translational movements.

Gasket

A gasket is a metallic plate positioned between the diamond anvils. Its primary role is to serve as a chamber for the sample. The gasket is also crucial for determining the highest pressure achievable within a diamond anvil cell. Common gasket materials include tungsten, rhenium, stainless steel T-301, martensitic tool steel, and Inconel.

Before loading the sample, pre-indentation of gaskets is performed by plastically deforming a metal gasket between the diamonds, forming a supporting ring and extruding outwards. This process is vital for preventing premature failure of diamonds and increasing the attainable pressure range. The hole drilled in the pre-indented gasket functions as the sample chamber, where pressure pushes outwards on the wall of the gasket hole. When the gasket is stable, an equal and opposite force acts inwards due to the shear strength of the gasket material and friction. The maximum pressure achievable is determined by the shear strength of the gasket material, its pre-indented thickness, and the size of the culet of the anvil. Typically, the size of the hole is half to one-third the culet diameter.

Pressure Transmitting Medium

To ensure hydrostatic conditions without differential stress on the sample, it is immersed in a pressure-transmitting medium. One of the most frequently used media is a 4:1 methanol-ethanol mixture, which provides quasi-hydrostatic conditions up to 10 GPa. Care must be taken when loading volatile liquids to prevent bubbling or evaporation before pressure is applied.

Pressure Calibration using Ruby Fluorescence

The ruby fluorescence technique is the most common method for in-situ pressure measurement in a diamond anvil cell (DAC). This method is favoured because ruby (Al_2O_3, Cr^{3+}) does not chemically react with most samples, and tiny ruby chips, as small as 1 μm, can be used without affecting the working volume of the sample. This allows for the placement of several tiny ruby chips in different regions of the DAC to determine pressure non-homogeneity.

2.2 Visit to Indus synchrotron facility for HP-XRD

X-ray diffraction is the most commonly used technique in high-pressure experiments for structural study. Since, the sample size is very small in DAC, brighter light sources like synchrotron is used for high pressure structural studies. Our team was granted **seven days of experimental access** to the '**Extreme Condition XRD Beamline**' **at RRCAT, Indore**. I have got an opportunity to visit the Indus-II angle dispersive XRD beam line at RRCAT. I cleared the safety training protocol for synchrotron beam use and participated in some of the high pressure XRD studies.

3. **Results and discussion of the sample studied**

- **Sample: $Na_2Cu_2TeO_6$ was selected as the sample compound for this study.** This choice was motivated by ongoing research in lithium-ion batteries, where efforts are directed towards developing alternative, more cost-effective, and reliable compounds through the substitution of lithium with sodium.
- **Procedure:**A diamond anvil cell (DAC) with a 350 μm culet size was used, along with a stainless-steel gasket of 230 μm thickness.The gasket was initially indented to a thickness of 70 μm.Subsequently, a through-hole with an approximate diameter of 150 μm was created at the centre of the indented gasket plate using an Electric Discharge Machine (EDM).The hole position was verified using a microscope. The gasket wascleaned with a sonicator.The gasket was affixed to the diamond using a soft adhesive.Subsequently, ruby powder, serving as a pressure

calibrant, was introduced into the cell using a sharp needle. This was followed by the addition of a pressure-transmitting medium, 4:1 methanol-ethanol mixture.The diamond anvil cell was then assembled.Then alignment of the Raman spectroscopy setup was carried out. We have performed high pressure measurements up to ~ 12 GPa. Following data acquisition, the measurement data wereanalysed using *Origin* software.

- **Results:**

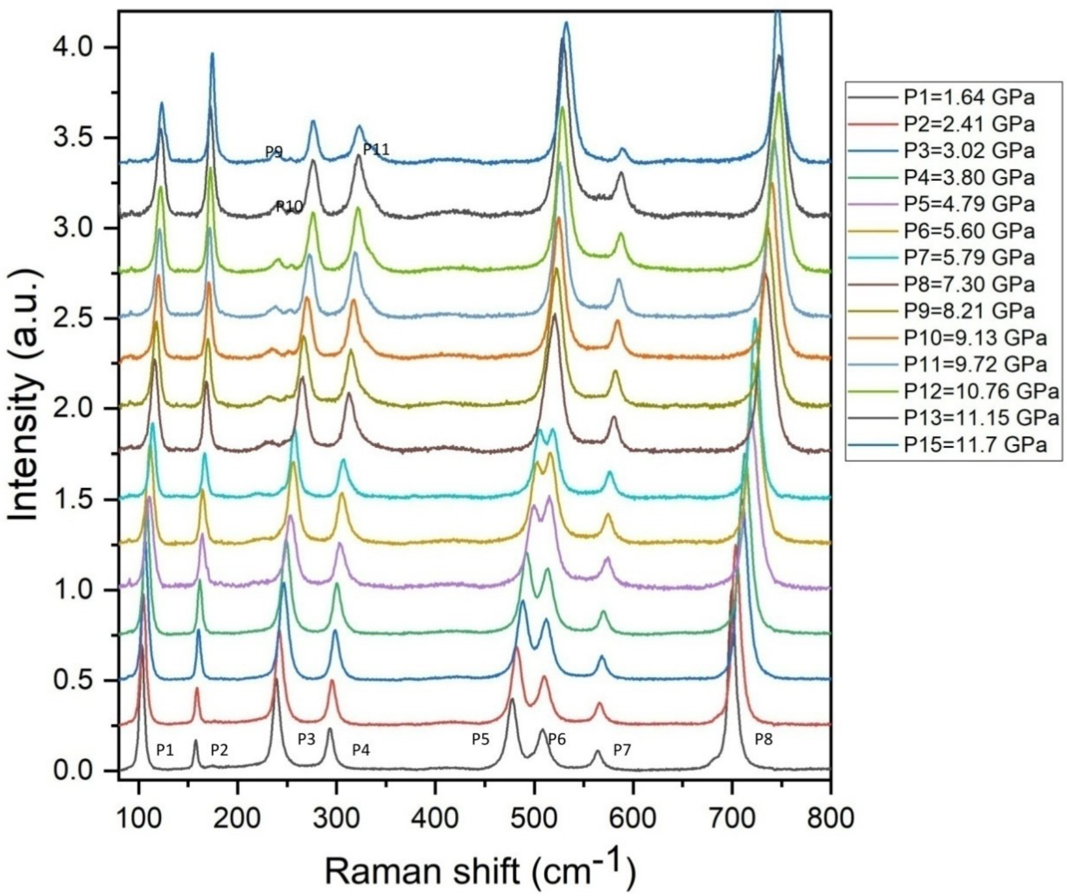

Figure 1. Pressure evolution of Raman spectra in Na2Cu2TeO6.

Figure 1 illustrates the evolution of the Raman spectra under varying pressure conditions. With increasing pressure, we observe variations in peak intensity of P1 and P2 peak. Specifically, the intensity of the first two characteristic peaks exhibits an inverse relationship: the intensity of the second peak increases while the intensity of the first peak decreases with increasing pressure.This observation suggests a spectral weight redistribution of the phonon modes.

Furthermore, two distinct peaks (P5 and P6) located around 500 cm^{-1}merge at an approximate pressure of ~7GPa. The emergence of three new peaks at this pressure denoted as P9. P10, P11 further suggests the possibility of a pressure-induced phase transition. To confirm this potential phase transition, further high-pressurestructural studies arenecessary.

Acknowledgement

I Jayesh Jadhav student of MSc physics at IIT Bombay successfully completed two-month long summer internship/training program at BARC. I worked and studied in high pressure and synchrotron radiation physics division from 10th may to 20th July.

I want thank physics division head Dr. T. Sakuntala mam for giving me this opportunity to work in one of biggest division at BARC, Physics group.Also, I would also like to thank Dr. Alka Garg for her teaching and checking my progress time to time. I am really grateful that she provided me with chance to accompany team visiting synchrotron at RRCAT Indore. Also, it is irresistible to mention snacks mam used to bring for everyone in lab.

Special thanks to my guide Dr. Subodha Sahoo for including me in the experiment,and for giving me insight into work along with real field research work. I thank him for sharing his valuable experience of PhD and of working as scientific officer at BARC with me.

I would like to thank Amit sir and Sathis sir for their time to time advices and helping me understand experiments.

Thanks to so many great people I had best time learning and socializing with everyone.

Tel.: +91-22-69299298
Fax: +91-22-25592183
Email: subodhas@barc.gov.in

सत्यमेव जयते

पूर्णिमाप्रयोगशालाएँ,
PURNIMA Laboratories,
ट्रॉम्बे,मुबई– 400085
Trombay, Mumbai – 400085

भारत सरकार Government of India
भाभा परमाणुअनुसंधान केंद्र Bhabha Atomic Research Centre
भौतिकी वर्ग Physics Group
उच्च दाब एवं सिंक्रोट्रोन विकिरण भैतिकी प्रभाग High Pressure & Synchrotron Radiation Physics Division

Ref.: HP&SRPD/Cert/2025 July 18, 2025

प्रमाणपत्र / CERTIFICATE

This is to certify that the project work entitled **"High pressure Raman spectroscopic study on Copper oxide systems"** is an authentic record of the project work submitted by **Shri Jayesh Janardan Jadhav**.

It is a faithful record of bonafide project work carried out under the supervision and guidance of **Dr. Subodha Sahoo**, Scientific Officer/E at High Pressure & Synchrotron Radiation Physics Division (HP&SRPD) of Bhabha Atomic Research Centre (BARC), Mumbai from 10.05.2025 to 18.07.2025.

18/07/2025

(Dr. Subodha Sahoo)
Scientific Officer/E

वैज्ञानिक अधिकारी/Scientific Officer
उच्च दाब एवं सिंक्रोट्रॉन विकिरण भौतिकी प्रभाग
High Pressure & Synchrotron,
Radiation Physics Division,
भारत सरकार, भाभा परमाणु अनुसधान केंद्र
Govt. of India Bhabha Atomic Research Centre,
ट्रॉम्बे, मुंबई/Trombay, Mumbai-400085.